Voice / Video over LTE (VoLTE / ViLTE)

Course objectives:

The course provides a solid technical understanding of the different LTE voice and SMS solutions endorsed by 3GPP, and the LTE industry. More specifically, the course covers three ways of providing Voice/SMS services over LTE. The non-IMS based solutions 'CS Fallback' and 'SMS over SGs', the IMS-based solutions 'Single Radio VCC' and 'SMS over IP' as well as VoLTE via Generic Access. The course contains an overview of LTE/EPC and IMS, targeted at the telephony services, so participants grasp a solid know-how on the technical aspects of the VoLTE.

Duration Instructional methods:

Option 1: On-Site Training: 5 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 5 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 10 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

Radio and Packet Core Engineers, Managers, Project Managers, IP Engineers, Technical Product Managers, Technical Marketing Officials, Technical Strategy Officials.

Voice / Video over LTE (VoLTE / ViLTE)

Course Contents:

- EPS Network Architecture description
  - LTE Network Architecture and Interfaces
  - EPC Network Architecture

- Circuit Switched Fallback (CSFB)
  - Combined PS/CS registration
  - The SGs-interface and the SGsAP protocol
  - CS Fallback call setup
  - Fallback options
  - Handover
  - Redirect
  - NACC

- The IP Multimedia Subsystem (IMS)
  - IMS network elements
  - The SIP and SDP protocols
  - IMS control of bearers/sessions
  - IMS application servers

- VoLTE Requirements and Protocols
  - SIP
  - SDP
  - RTP and RTCP
  - Robust Header Compression (ROHC)
  - Media (voice and video) codecs and compression

- End-to-end signaling flow discussion
  - UE Registration for VoLTE / ViLTE
  - VoLTE call scenario
  -
- Single Radio Voice Call Continuity (SRVCC)
  - The Sv-interface and the enhanced MSC Server
  - Service Centralization and Continuity AS (SCCAS)
  - The GSMA VoLTE and LTE roaming initiatives
  - SRVCC handover to GERAN/UTRAN

- Solutions for SMS over LTE
  - SMS over Sgs-interface
  - SMS over IP and the IP-SM-GW
  - Transport-level vs service-level interworking

- VoLTE Services
  - MMTel Supplementary Services
  - Service-related P-headers
  - IMS Communication Service Identifier (ICSI)
  - XCAP over the Ut interface
  - DTMF
  - Emergency calls

- Security Mechanisms
  - Re-using mobile network attach
  - IMS-AKA

Pre-requisites:

Basic knowledge of the LTE/EPC and legacy 2G/3G systems (GSM/UMTS), IP-based communication.

Course Code: VoLTE

Price per student:

For Option 1, Classroom training : USD 2,400.00 per student

For Options 2, 3, Virtual Classroom training: USD 1,900.00 per student